# PIZZA SALES DATA ANALYSIS

*Detailed Project Documentation*

**Independent Data Analytics & Business Intelligence Project**

## 1. Executive Summary

This independent project demonstrates an end-to-end data analytics workflow using pizza sales transaction data. The project moves from transactional data understanding and validation to SQL Server analysis, KPI calculation, Power Query transformation, DAX measures and an interactive Power BI dashboard. The purpose is not only to calculate metrics, but to translate sales data into business-oriented insights that can support inventory planning, staffing, promotions, menu optimization and product strategy.

## 2. Project Objective

The main objective is to build a structured sales analysis solution that provides management with a clear view of revenue, order activity, pizza volume, customer ordering behaviour and product performance.

- Calculate core sales KPIs and establish an overall performance baseline.
- Analyse order patterns across days and months.
- Understand sales contribution by pizza category and size.
- Identify top- and bottom-performing pizzas using revenue, quantity and order metrics.
- Present the analysis through an interactive Power BI dashboard.
- Convert analytical results into practical business recommendations.

## 3. Business Problem & Analytical Questions

The business has historical pizza sales transactions, but raw transaction-level data does not directly provide an easy management view. The project therefore answers the following questions:

- What is the total revenue generated?
- How many unique orders were placed?
- How many pizzas were sold?
- What is the average order value?
- How many pizzas are sold per order on average?
- Which days and months generate higher order volumes?
- Which pizza categories and sizes contribute most to sales?
- Which pizzas are the strongest and weakest performers by revenue?
- Which pizzas have the highest and lowest quantities sold?
- Which pizzas appear in the highest and lowest number of orders?

## 4. Dataset & Important Fields

The analysis uses a pizza sales transaction dataset. The project uses the following fields in its SQL and Power BI analysis:

| Field | Role | Description |
|---|---|---|
| order_id | Order | Identifier used to count distinct customer orders. |
| order_date | Time | Date used for daily/monthly trend analysis. |
| order_time | Time | Transaction time available for time-related analysis. |
| pizza_id | Product | Identifier for the pizza item. |
| pizza_name | Product | Pizza name used for product ranking. |
| pizza_category | Category | Category used for category-level sales analysis. |
| pizza_size | Dimension | Size used for sales contribution analysis. |
| quantity | Measure | Number of pizzas sold. |
| unit_price | Measure | Price per pizza. |
| total_price | Measure | Revenue generated by the transaction. |

## 5. End-to-End Project Workflow

The project follows a complete analytics pipeline:

- Raw Dataset
- Data Understanding
- Data Cleaning & Validation
- SQL Server Database
- SQL Analysis
- KPI Calculation
- Power Query Transformation
- DAX Measures
- Power BI Dashboard
- Business Insights
- Business Recommendations

## 6. Data Preparation & Validation

- Reviewed the dataset structure and the meaning of the transaction fields.
- Checked data types, particularly date, numeric and identifier columns.
- Checked for missing values and duplicate records.
- Validated order identifiers and product/category fields.
- Validated numerical fields used for aggregation.
- Checked revenue logic using quantity and unit price.

The objective of this stage was to make the dataset reliable enough for aggregation and visualization. The project documentation does not claim a specific number of missing or duplicate records unless that result was explicitly calculated in the analysis.

# 7. SQL Server Analysis

SQL Server is used as the analytical layer for the project. The SQL analysis is organized into four major areas: KPIs, time-based analysis, category/size analysis and product performance.

## 7.1 KPI Analysis

- Total Revenue
- Total Orders
- Total Pizzas Sold
- Average Order Value
- Average Pizzas Per Order

The project uses SUM(total_price) for revenue, COUNT(DISTINCT order_id) for unique orders, SUM(quantity) for pizzas sold, and division-based calculations for average order value and average pizzas per order.

## 7.2 Time-Based Analysis

Daily analysis groups orders by weekday using DATENAME(DW, order_date). Monthly analysis groups orders by month using DATENAME(MONTH, order_date). Distinct order IDs are used when counting orders so that multiple pizza lines belonging to the same order are not incorrectly treated as separate orders.

## 7.3 Category & Size Analysis

- Percentage of sales by pizza category.
- Percentage of sales by pizza size.
- Total pizzas sold by pizza category.

Percentage-of-sales queries compare each category or size's revenue contribution with total revenue.

## 7.4 Product Performance Analysis

- Top 5 pizzas by revenue.
- Bottom 5 pizzas by revenue.
- Top 5 pizzas by quantity.
- Bottom 5 pizzas by quantity.
- Top 5 pizzas by total orders.
- Bottom 5 pizzas by total orders.

Using separate revenue, quantity and order rankings avoids treating one performance dimension as a substitute for another. A pizza can have high order frequency without being the highest-revenue product, so these metrics provide complementary perspectives.

# 8. Power BI Development

The SQL Server data is brought into Power BI for transformation, modelling, calculation and visualization. Power Query supports data preparation and DAX provides reusable measures for KPI calculations.

### 8.1 Core DAX Measures

```
Total Revenue = SUM(pizza_sales[total_price])

Total Orders = DISTINCTCOUNT(pizza_sales[order_id])

Total Pizzas Sold = SUM(pizza_sales[quantity])

Average Order Value = DIVIDE([Total Revenue], [Total Orders])

Average Pizzas Per Order = DIVIDE([Total Pizzas Sold], [Total Orders])
```

### 8.2 Dashboard Components

- KPI cards for the main sales measures.
- Daily order trend.
- Monthly order trend.
- Revenue contribution by pizza category.
- Revenue contribution by pizza size.
- Top 5 and Bottom 5 product analysis.
- Interactive filters/slicers for relevant dimensions.

## 9. KPI Definitions

| KPI | Calculation | Business Meaning |
|---|---|---|
| Total Revenue | SUM(total_price) | Overall sales value generated. |
| Total Orders | COUNT(DISTINCT order_id) | Number of unique orders. |
| Total Pizzas Sold | SUM(quantity) | Total pizza units sold. |
| Average Order Value | Total Revenue ÷ Total Orders | Average revenue generated per order. |
| Average Pizzas Per Order | Total Pizzas Sold ÷ Total Orders | Average number of pizzas in an order. |

## 10. Business Insights Framework

The dashboard is designed to support business interpretation rather than only reporting numbers. The following insight areas are central to the project:

**Time-based performance:** Identify high- and low-demand days and months to support staffing, preparation and inventory planning.

**Product performance:** Identify products that contribute strongly or weakly to revenue, quantity and order volume.

**Category performance:** Understand which pizza categories contribute most to revenue.

**Size preference:** Understand the relative sales contribution of different pizza sizes.

**KPI movement:** Use the core KPIs as a baseline for tracking sales performance over time.

## 11. Business Recommendations

**Inventory Planning:** Prioritize ingredients and preparation capacity for products and periods with stronger demand.

**Staffing:** Align staffing levels with periods showing higher order volumes.

**Promotions:** Use targeted promotions to improve visibility or trial for low-performing products.

**Menu Optimization:** Review products that consistently rank near the bottom across relevant metrics.

**Product Strategy:** Focus marketing and merchandising on products with strong revenue or demand.

**Seasonal Planning:** Use monthly patterns to prepare inventory, staffing and promotional activity.

## 12. Technical Skills Demonstrated

- SQL Server: querying, aggregation, GROUP BY, ORDER BY, DISTINCTCOUNT-equivalent logic, CAST, DATENAME, TOP and subqueries.
- Data Analysis: KPI definition, trend analysis, contribution analysis and product ranking.
- Power Query: data preparation and transformation.
- DAX: reusable business measures and KPI calculations.
- Power BI: dashboard design, visualization, filters and interactive analysis.
- Business Analysis: converting business questions into analytical requirements and recommendations.

## 13. Project Deliverables

- Pizza_Sales_Analysis.sql — SQL analysis queries.
- Pizza_Sales_Dashboard.pbix — Power BI dashboard.
- Dashboard screenshots — portfolio visuals.
- Pizza_Sales_Project_Documentation.docx — detailed project documentation.

## 14. Portfolio / GitHub Structure

The project is being organized as a portfolio repository so recruiters can review both the final dashboard and the underlying analytical work.

- README.md — project overview, objectives, workflow, KPIs and links.
- SQL — SQL Server analysis queries.
- PowerBI — Power BI dashboard file.
- Screenshots — dashboard images.
- Documentation — detailed project documentation.

## 15. Project Limitations

- This analysis is based on a **generic pizza sales dataset** and is not associated with or representative of any specific pizza company or organization.
- The dataset is used for **data analytics and portfolio demonstration purposes** rather than evaluating the actual performance of a particular business.
- The findings and recommendations are intended to demonstrate the **data analysis methodology and technical skills** rather than represent current company-specific business decisions.

## 16. Conclusion

This project demonstrates an end-to-end approach to data analysis: starting with transactional data, structuring and analysing it in SQL Server, transforming and calculating measures in Power BI, visualizing performance through a dashboard, and translating the results into business-oriented recommendations. It demonstrates practical skills relevant to Data Analyst, BI Analyst and Business Analyst roles.

## 17. Personal Learning Outcomes

- Improved ability to translate business questions into SQL analysis requirements.
- Strengthened SQL aggregation and analytical query skills.
- Developed practical experience connecting SQL-based analysis with Power BI.
- Improved understanding of KPI design and DAX measures.
- Learned to communicate analytical results as business insights and recommendations.
- Built a portfolio-ready independent analytics project.